

Reality

By Jason P. Hayward

Table of Contents

[Introduction](#)

[Disclaimer](#)

[Part I: Filter](#)

[I.1: Sources of truth](#)

[I.2: Values](#)

[I.3: Success](#)

[I.4: Hypotheses](#)

[I.5: Confirmation bias](#)

[Part II: Intake](#)

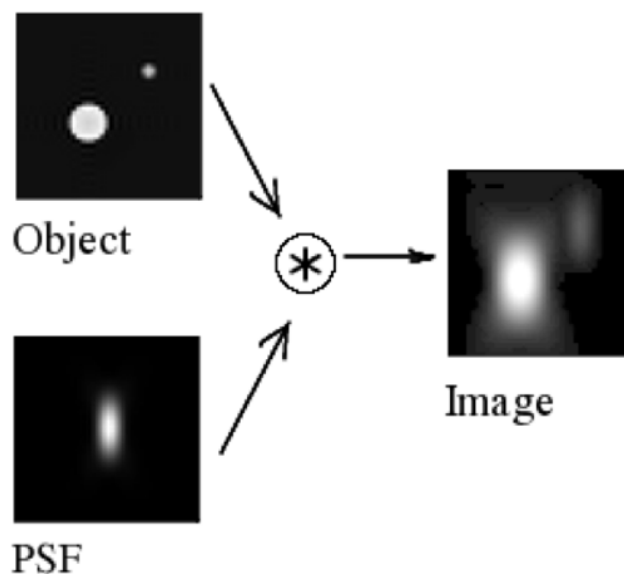
[Part III: Reality](#)

[Appendix: my view](#)

Introduction

I am a professor of engineering, and my speciality is radiation sensing. Of course, we don't naturally sense radiation (or 'see' it), so we create instruments to make this kind of perception possible, similarly to the way eyes were created to see and ears were created to hear. When it comes to our sight, we don't see things as they are. Instead, we see them through the instrument of our eyes, and eyes don't all see the same way.

The way this is explained in the field of imaging is shown in the figure below. In the top left illustration labeled as object, there are two circular shapes, one larger and one smaller, which represent the true form of an example object. The response of the instrument of vision (for example, one's eyes or a camera of some sort), is shown at bottom left as a point spread function (labeled as PSF). The instrument response has a blur associated with it, which intrinsically will cause some sort of smearing of the appearance of the object. The image that is observed (at right) is produced via a 'folding' or 'convolution', represented as $*$, of the object with the instrument response function. The result is a blurry image that bears some resemblance to the original object, albeit imperfect. Further, if one starts with the image and the instrument response is well known, it is possible to work backward, from right to left, to 'unfold' or 'deconvolve' that response from the image, which, if done effectively, yields a very good estimate of the appearance of the original object.



Explaining this process of vision bears some similarity to the means by which we perceive reality. Reality as we see it could be compared to the resulting image. Reality as we perceive it has been impacted mainly by two things: 1) the information we have taken in and the things we have experienced throughout our lives, and 2) the filter of our mind that shapes our view of the information we have taken in and the things we have experienced. The former could be compared to the object viewed, while the latter could be compared to the point spread function

(or instrument response). For simplicity, the first will be called our intake, while the second will be referred to as our filter.

As such, this book about reality as we see it is separated into three parts: filter, intake, and reality. It considers the way different people see the world we live in, affected by the sources of truth we choose to accept, those things which interest us, the biases we take on, and also spiritual life. Hopefully, it leads us to step outside of ourselves, as best we can, to reflect and think critically on how we've arrived at the point where we are, and also to consider where we are headed.

I'm excited about this! I hope you are as well.

Disclaimer

Just as everyone has a filter, so do I! First, I trust the Bible as the word of God, inspired by the Holy Spirit. Second, I trust science, particularly the study of nature, as a source of truth about the world.

Since the former is more broadly accepted than the latter in today's world, here are some reasons I find it reasonable to believe the 66 books of the Bible as God's Word:

- 1) The Bible has been a strong force for freedom and the worth of each human, lifting up the dignity of every race, both men and women, slave and free, and every form of marginalized person
- 2) Most of Jesus' disciples and apostles both lived and died for the gospel including the truth of the resurrected, living Jesus; many saints in the early church did the same
- 3) The truth of the miracles and healings recorded in the Bible in Jesus' Name is more believable in that testimonies of the same continued afterward and continue to this day; there are way too many credible ones to refute; the belief that the natural is all there is completely without merit
- 4) The testimonies of people's lives who have been changed in beautiful ways by making Jesus Lord of their lives and actually following Jesus despite the cost
- 5) The gospel was spread into the world by the power of God (healings, miracles, signs), the blood of the martyrs of the church, and the testimonies of changed lives
- 6) The prophecies of Jesus in the Old Testament were fulfilled in the New Testament by Jesus of Nazareth, and the overall narrative of God's relationship with mankind in the old covenant points toward Jesus
- 7) The Bible is historically accurate and has been verified again and again by historians and archeologists; it contains no contradictions (in the strict sense of the word)
- 8) The Bible is the most read and most famous book in the world
- 9) The Bible is good news, fulfilling our most basic needs to be deeply and unconditionally loved by the God that created us and to belong to a community for which we were made

I also think there is some truth to be found in the other sources listed in the next chapter: our five senses, our experience, history, media, and tradition. Regarding other sources of special revelation outside of the Bible, I believe that God still speaks through prophecy to this day, but we must be careful to test prophecy, holding fast to what is good (1 Thessalonians 5:20-21). In particular, it should be consistent with the nature and the revelation of God from the Bible.

My strongest values include faith, family, love, humility, commitment, and service. I've also been shaped by my life experiences, including the church communities I've joined, my academic studies, my military experience, and my current profession in academia and research.

Part I: Filter

I.1: Sources of truth

The way we filter what we take in depends upon what we have accepted as our sources of truth and how we've effectively ranked them in importance to us.

Five senses. We all tend to trust our five senses, unless they are not fully functioning. Our five senses, of course, are sight, hearing, touch, taste, and smell. They are the first means we are able to use to learn about our world. Knowledge that comes to us via our senses has been called sense knowledge.

Experience. Some of us may value our own experience as the greatest source of truth. Our experience does rely on our own sense knowledge. We may also value the experience of others, especially loved ones, that we may hear by way of storytelling. The history of others that we hear comes through some filter, usually the filter of the person relating their experience.

History. History is the learned experience of others that have come before us, as relayed by a particular source. As most of us have heard, if we don't learn from history, we are doomed to repeat it (attributed to George Santayana). The history that we hear has already come through some filter, so one learns that one must decide which sources are trustworthy by way of their own filter.

Media. Unless one lives in a cave, we receive news about the world and the happenings of the world around us with some frequency, most often of our choosing. The news we hear has also already come through some filter. Entertainment media includes human interest stories, movies, television shows, or celebrity news. Media is frequently designed to influence public opinion. Social media is often more personal in nature, distributing filtered information to an audience of followers.

Tradition. Some religious groups take their religious customs or beliefs of the prior generations that preceded them as a source of truth regarding what is right and proper. In the time of Jesus of Nazareth, he referred to this source of truth as the "traditions of men," according to the gospels of Matthew and Mark. Tradition is filtered via those from whom we have learned it.

Nature. Many look to the natural world as a source of truth. Scientists are certain among those who do. Some religious groups have likewise referred to nature as "the book of nature" in the way that it reveals truth. Nature is a source of truth that we can take in on our own, or we can look at it filtered through the interpretation of others.

Special revelation. Special revelation has often been said to come from prophets recognized by certain groups; these are prophets that speak God's words or have recorded God's spoken words. The most influential sacred texts in history, some of which are attributed to prophets, include the Bible; the Tanakh (which includes the Torah), the Talmud; the Quran, the Hadith; the

Vedas, the Upanishads, the Puranas, the Bhagavad Gita; the Tripitaka, the Buddhist Sutras; and the Tao Te Ching, the Zhangzi, and the Daozang. We are able to read these sources that may be or may become revelation of truth to us on our own, or we can learn about them filtered through others, such as teachers, preachers, and commentators.

How are these sources ranked to you? Why?

1.2 Values

What do you value?

The values we take on in life affect the nature of our filter that we use to process and see our world.

Our values are shaped by our society, our upbringing, our experiences, and by our faith or worldview. Our values affect what we think about, what we desire, and the interests we seek to affect in our decision making.

When it comes to our relationships with others, some value **the family unit**, one man married to one woman for life, that may procreate children and raise them up together. Some extend the definition of the family unit outside of the conventional marriage of one man to one woman. Some value **strong extended families**, especially to include aging parents, which they take care of in their own homes, and who help to raise and take care of the children as they grow.

When it comes to relationships outside of family, some greatly value **friendship**, treating friends as some would their nuclear family. Some greatly value **community**, especially local community. Some may value the community of a religious fellowship. Values related to community may include **teamwork**, **collective responsibility**, **consensus**, and **unity**. **Nationalism** extends great value to those of one's nation. Related values may include **national teamwork**. Some may value **deference to authority**. A spirit of **patriotism** tends to always favor the best interest of one's own nation.

Moving more toward the individual rather than a community, some may greatly value **individualism**, personal **liberty**, **equality**, **personal responsibility**, **discipline**, **intelligence**, **creativity**, and **work ethic**. Some may value **educational achievement**. Some may value **free enterprise**.

Others value particular virtues including **love**, **peace**, **courage**, **compassion**, **kindness**, **generosity**, **honesty**, **integrity**, **authenticity**, **loyalty**, **justice**, **humility**, **moderation**, **perseverance**, **commitment**, **tolerance**, **patience**, **chastity**, **contentment**, **service** and **faith**.

How are these values ranked to you? Why?

I.3: Success

What do you believe success is?

What does success have to do with family? Are there people in your family that you would like to think well of you? What is your hope for your family and your family relationships? Are you hopeful to leave a certain legacy to your family? How is your success linked to the achievements of your family?

What does success have to do with your educational achievement? Have you achieved what you set out to achieve? Or what is your hope?

What does success have to do with work? What does it have to do with your performance and achievement in the workplace? What is your hope for achievement through work and your career? Are you hopeful to have a particular impact or leave a particular legacy through your accomplishments at work?

What does success have to do with your friends, your community, or your nation? Do you have the friends and relationships outside of family that you would like to have, or what is your hope? How do you want your friends or your community to think of you? Are you hopeful for your community to meet certain goals or outcomes? Do you desire to have greater influence?

What does success have to do with where you live and what you own? What does it have to do with the opportunities available to you? What do you hope for in the future?

What does success have to do with your character and the ways you demonstrate that character? Do you hope for growth as a person?

What does success have to do with your faith? Do you desire to please God? If so, what is it about your life that you expect will please God?

We usually orient ourselves toward and head in the direction of what we consider to be success. Where are you headed?

How does your definition of success alter your filter for how you see the world and the people in it?

I.4: Hypotheses

We all have a set of hypotheses in our mind about the way things are. It's our working model.

How did the world we live in come to be? How did life come to be?

Is there life beyond our planet? Are there dimensions beyond which we cannot or have not yet experienced?

Is the supernatural real, or is the natural world all that there is? Are supernatural powers real? Are there miracles?

Is there a God? Can we know things about God? If so, what do we know about the character of God?

What is God's involvement in the world we live in, if any? Does God get involved in our personal affairs?

Will this God judge us one day? Are heaven and hell real? Who goes to heaven?

Is evil real? If so, what is the nature of evil?

What is right? How would you describe common morality? What's the nature of common morality?

Is truth relative or absolute?

What is your place in the world? How important are you? How important are other living things?

What value do other people have? How does one's value depend on who they are or what they have accomplished?

Do we have purpose in this world? If so, what is it?

I.5 Confirmation bias

One of my colleagues first introduced me to a paper on confirmation bias¹, and it makes a lot of sense. Confirmation bias connotes the seeking or interpreting of evidence in ways that are partial to existing beliefs, expectations, or a hypothesis in hand. Real-world contexts presented in that article included number mysticism, witch hunting, policy rationalization, medicine, judicial reasoning, and science were presented. In view of the evidence, it was concluded that confirmation bias is pervasive and strong.

This idea of confirmation bias relates to this idea of the filter that we have each formed to shape our perception of the world. We each have a working model we've put together in our minds of the way things are. This working model includes existing beliefs, expectations and hypotheses that we have.

Do you think confirmation bias applies to you and how you may view new experiences and new information? Why or why not?

¹ R.S. Nickerson, Confirmation Bias: A Ubiquitous Phenomenon in Many Guises, Review of General Psychology, 1998, Vol. 2, No. 2, 175-220.

Part II: Intake

Our way of taking in information from the world we live in has changed drastically in the last hundred years! Radio started in 1920, in Pittsburgh, PA. In 1925, the first television signal was transmitted. Commercial TV broadcast began in 1947. The first geosynchronous communications satellite was launched in 1963. The first modern fax machine became commercially available in 1964. The first email was sent in 1965. The personal computer market was born in 1976. Mobile phone service started in 1981. In 1985, America Online internet service was launched. In 1989, the world wide web was prototyped at CERN. Texting began in 1992. Internet radio broadcasting started in 1994. Skype video calling started in 2003. Facebook was launched in 2004, YouTube began in 2005, and the iPhone first rolled out in 2007.

Today, people mainly take in information from their personal devices. Globally, people average about 7 hours of screen time each day for internet-connected activities, much of it on mobile phones, including about two and a half hours of social media, an hour and a half of streaming music, an hour of listening to podcasts, and an hour playing games. More than half of us get our news from digital devices. On digital devices, we get news from news websites or apps, search engines, social media, and podcasts.

How do you take in information now? How do you decide what you do and do not believe?

What type of information do you take in most? Is it affected by your filter?

If you are an adult, how is this different from how it was before you turned thirteen? In your teenage years?

Why do you think different people tend to see and experience the same world so differently than you do?

How do you think your personal experience has contributed to the ways you have come to think and view the world? What about the experience of growing up in your family?

What about your experience in being a part of the communities you have known? What communities and groups have influenced who you are today the most?

What key opportunities that you've had in life have influenced who you are today?

How has schooling influenced you? If you've had a choice in where you have gone to school, why did you choose what you chose? How do you think this choice (or these choices) influenced you?

With whom do you feel safe?

Who has been your intellectual community?

Have you had mentors outside of your family that have had an influence on who you are today? How so?

Who have your role models been? Why do you think they became your role models?

Have you ever tried to specialize in some area of study or performance, affecting the focus of your intake? Most of us do in some way.

Becoming excellent in any area of life typically takes both mentorship by a good teacher and practice. In particular, olympians who move on to excel are most often trained by former top olympians. The best soldiers are trained by good soldiers. Likewise, researchers who excel are trained by successful researchers who are good teachers.

In Malcolm Gladwell's book *Outliers*, he proposed that it takes 10,000 hours of intensive practice to achieve mastery of complex skills and materials, like playing the violin or getting as good as Bill Gates at programming. Concerning practice and habits, some psychologists have said that a minimum of 63 days of repetition is needed by people in order to break a habit or to form a new habit. In the case of becoming an olympian, the training process typically takes four to eight years.

Sometimes this strict training one receives, such as the mental and physical training of those who enlist in military service, might even be compared to indoctrination. The end result is seen by many as laudable. In addition to physical training, recruits are instilled with an understanding of, and willingness to live by, the values held by each service. In the process, civilians are changed into part of something larger, a collective group that shares a unique identity. In the process, former personal characteristics detrimental to military life are removed. Individuals may be trained to kill when necessary, and they are trained to view themselves in collective terms. Bootcamp on its own lasts up to 13 weeks (or 91 days), but training continues after that point. For example, if one wants to train to become a sharpshooter, it could take several more months of intense training to gain the necessary skills and knowledge.

What do you think about such strict training? What do you think about indoctrination?

When one submits to an educational system to be trained as a researcher, one is trained how to think critically, how to write, and how to speak, often over a period of four or more years. The training takes the form of practicing research on one or more research projects while being mentored in the work by one or more senior researchers. One learns to read the scientific literature in their discipline, think critically about it, and then make original contributions to it. When one finally obtains their terminal degree, often a Ph.D., then one has the opportunity to join the discipline, or the community of researchers, in an official capacity.

How have you submitted yourself to learn from trusted others?

Part III: Reality

Veritas means truth or reality in latin.

What is truth? What is reality?

How do you decide what new information you receive is true and what is false, or fake?

For each of us, our view of reality is formed over time based on our intake and observations, our interpretation of them via our filter, and the way we've made sense of what we take in and observe in the context of the model of life that we've put together in our mind.

It's essentially our set of estimations and assumptions that we've made of the way things actually are. It's our prediction of "ground truth."

This might be called our worldview. We each have one.

It includes our views of theology, philosophy, ethics, biology, psychology, sociology, law, politics, economics and history. By it we make sense of God or the absence of God, the natural world we see, morality, life, consciousness, our relation to other people, government, the rule of law, how we should spend money and resources, and the past.

How did you come to believe what you believe today?

Is it right?

In science and engineering, models are often used to interpret new results and also to predict what we can expect to come next. For example, in weather forecasting, one cannot collect data from the future, so the models have to use estimates and assumptions to predict future weather, such as hurricane trajectories. Yet, weather is affected by many variables; the atmosphere is chaotic, changing all of the time; observation networks are limited in space and time; and all observations have uncertainty associated with them, so those estimates are less reliable the further one gets into the future. When it comes to hurricane forecasting, the final accuracy of models at the time evacuation announcements need to be made is not great, but greater accuracy has been found when forecasts from a collection are combined together.

In the end, when it comes to complex phenomena such as hurricanes, no model is right on.

Reality in its totality is also like this. It's complicated.

At any time, we may think we have it all figured out. But, we don't. We may not even be close.

The automobile drivers of the world are nearly united in just one thing: nearly all of us think we are above average drivers. Clearly, that's not possible. Do you think it could also be the same

for nearly every society that has ever lived, in that we think we may be the smartest and most capable generation that has ever lived?

Do we keep on getting better? Do we continue to improve our ability to arrive at estimates of what's real and true?

In science, we tend to find in time that our interpretations in certain areas have been wrong. Assumptions or estimates had been incorrect. There is significant disagreement about those considered to be the best in the world.

In interpretations of religious texts, there is similar difficulty. Interpretations can be wrong. Great minds often do think differently.

You can bet that you are wrong about certain things— probably some that are pretty big. I can and should say the same.

So the end result is, it's time to have some humility, whoever you are. Let's recognize our limitations as observers in this large, complex, multi-faceted world. It's unnatural to have all the truth.

Also, let's not overestimate our capability to precisely understand what is true and what is real.

But, let's continue to pursue it, and let's be open to see what we can learn from each other.

In the end, we are better together.

Appendix: my view

I encourage you to consider the questions in this book that are helpful to you and parts of this book you agree with, and learn as much about your intake, filter, and reality as you can. Also, where are you headed?

I consider this thought experiment next for me as an example for anyone who would like to keep on reading.

In my view, the main two sources of truth are the “book of nature” (Romans 1:20-21) and God’s book of revelation, the Bible (Matthew 5:17-19, 2 Timothy 3:16, 2 Peter 1:16-21). My influence in adding the book of nature to the top of my list can be most recently attributed to Hugh Ross, based on some personal reading I did some years ago.

In my view, reality is God, and what we know of Him is known through the Bible and through nature. My worldview has sometimes been called Biblical Christianity. Clearly, it’s theistic. As far as philosophy, I believe in the supernatural, including the tripartite God and Satan, angels and demons, signs and wonders. I think it’s quite likely that heaven is here in our midst, contained within extra dimensions beyond the four that we experience with our five senses. As far as my ethics, I believe in ethical absolutes defined by God’s words in the Bible.

As far as biology is concerned, I believe in creationism that is understood through a combination of interpretations of the revelation in the Bible and interpretations of scientists of the natural record. The nature of our origins is not precisely known, but the study of it is very interesting.

In psychology, I believe in a tripartite nature that includes spirit, soul and body. The spirit communicates with God, and the soul includes mind, will and emotions. In sociology, I believe in the value of the home, the church and the state. In the realm of law, I believe in both biblical and natural law. In politics, I believe in justice, freedom and order.

In economics, I believe both in stewardship of property and also that the government must intervene to ensure a minimum level of welfare for all citizens, because the people in private organizations are not collectively generous enough to ensure a state of welfare for all that is appropriate considering the great value of human life. I believe in recorded history, including the proven record of evolution and the historical resurrection of Jesus Christ.

I believe the greatest truth is revealed by the word of God in the Bible, which should be read literally and must be interpreted in such a way as to separate out what it says from the religious traditions of mankind. I believe that all human life has special value because we are all created in the image of God (Genesis 1:26-27). This means that as creations of His, meant to be His cherished family members, we are able to take on many of God’s personal characteristics, including His love, mercy, patience, and holiness. The person of Christ is the exact imprint of the image of God (Colossians 1:15, Hebrews 1:3). The word of God is the blueprint for what is right. The image of God was revealed in the person of Jesus Christ, who was the word of God made

flesh. He came into the world to fulfill the law and the plan of God motivated by His great love for us.

God's family of Spirit-filled Christ followers are called the church. They are Christ's living body in this world, indwelt by the same Holy Spirit of God that lived in Jesus Christ (Acts 10:38) and raised Him from the dead (Romans 8:11). The extended family of the church is the most important community in the world. God meant it for belonging, equipping, and also for carrying out the function of continuing the ministry of Jesus Christ that He started while on earth. The church is to stay united, and each member of it is to take on individual responsibility such that the church together fully takes on its collective responsibility of meeting the needs of the world. The purpose of the church is to stay aligned with God's will and to do God's will on earth as it is in heaven (Matthew 6:10). It is to be led by the Spirit of God (Romans 8:14), doing the word of God described in the Bible. Also, while it's unnatural to have all truth, the indwelling Spirit of God supernaturally guides us into all the truth (John 16:13).

In terms of the liberty that we have in Christ, we are set free to do God's word; to serve all in love (Galatians 5:13-14); and to do good to all, especially those of the household of faith (Galatians 6:10). All of the virtues listed in [1.2: Values](#) are of great value, especially faith, hope and love. The greatest of these is love (1 Corinthians 13:13). Love is not a feeling; it is benevolent, always *willing* and *taking action* to do good and to see things made right.

Success is pleasing God and bringing glory to God by obedience to all of His commands and by attention to do all of His will. It is accomplished especially through growing up into the fullness of the image of Christ in cooperation with God's indwelling Spirit, and by discipling others toward the same (Ephesians 4:11-16).

My intake in the past few years has included a lot of Bible and Bible study on digital devices, as I seek to renew my mind to the word of God (Romans 12:1-2, Ephesians 4:22-23, Colossians 3:9-17). It still includes media consumption, including internet television, social media (see [here](#) and [here](#) in particular; feel free to reach out, if you like), and some internet news. It also includes scientific and engineering literature and publications for work. The bulk of what I've contributed to at work in recent years is found [here](#) and [here](#). It also includes family and fellowship.

Where am I headed? There's great hope ahead! There is expectation of good based on the finished work of Jesus Christ. His Spirit lives in me, so as I behold the glory of the Lord, I can expect to be "transformed into the same image from one degree of glory to another" (2 Corinthians 3:18). I am also an ambassador for Christ (2 Corinthians 5:20), so that I get to see the same happen in the lives of others as I help disciple them. I have been seated "in the heavenly places in Christ Jesus, so that in the coming ages he might show the immeasurable riches of his grace in kindness toward [me] in Christ Jesus." Also, I am a worker together with God, and if the work that I have built on the foundation of Jesus Christ in this life survives the fire of God's testing, I can expect a reward from God (1 Corinthians 3:9-14).